

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

GLY 421 — Past Questions & Materials
by Renance

Course: GLY 421
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 400

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Paper 1 of 2 — GLY 421 PQ2 2024

Level 400 • Second Semester • PastQuestion

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Table 1: Strike and angle of dip measurements of cross-beddings associated with the sandstone outcrop

Strike	Dip (°)	Strike	Dip (°)
000	22 E	350	24 W
006	24 E	004	20W
356	20 E	008	22W
180	22 W	004	20 W
352	24 N	350	24 E
000	20 W	348	24 E
000	24 E	360	20 E
354	20 W	002	22 W
098	28 S	044	24 NW
096	24N	274	22 S
002	18 E	004	20 E
010	20W	000	18 W

Question 5:

5a). "The importance of basin analysis for Petroleum exploration cannot be overemphasised". Discuss. (7 marks)

5b). Use illustrations to show how named well-logs are applied for the prediction of a petroleum reservoir rock and its fluid contents. (5 marks)

5c) "Rocks generally keep their own records". Discuss the statement showing how such records are often read in sedimentary rocks (5.5 marks)

Question 6:

The results of Grain-size analysis are presented in Table 2.

a) Plot cumulative frequency curves for Samples A and B on the same Graph paper. (4.5 marks)

b) Compute and interpret: mean size, and standard deviation for samples A and B. (6 marks)

Table 2: Grain-size analysis carried out on some sandstone samples

S#	A	B	C	D	E
Sieve size (mm)	Weight retained (g)	Weight retained (g)	Weight retained (g)	Weight retained (g)	Weight retained (g)
4.000	0	0	0	0	0
2.828	0.025	0.025	0	0.025	0.025
2.000	0.025	0.025	0.5	0	0.025
1.414	1.95	1.95	2.5	1.95	0
1.000	13.0	15.0	12.00	13.0	15.0
0.707	16.0	14.0	13.0	16.0	12.0
0.500	12.5	10.5	13.0	12.5	12.0
0.353	5.5	7.5	7.5	5.625	5.5
0.250	0.9	0.9	0	0.7	0.9
0.177	0	0	1.5	0	2.00
0.125	0	0	0	0	2.00
0.0883	0	0	0	0	0
Pan	0.1	0.1	0	0.2	0.55

6a). Name any four (4) specific palaeoenvironmental indicators and their inferred environment of deposition. (4 marks)

6d). Using a ternary diagram ONLY, attempt the classification of sandstone on the basis of their framework composition of quartz, feldspar, and rock fragments (3 marks)

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